```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder

# Dataset
data = {
    'Outlook': [
        'Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast',
        'Sunny','Sunny','Rain','Sunny','Overcast','Overcast','Rain'
    ],

    'Temperature': [
        'Hot','Hot','Hot','Mild','Cool','Cool','Cool',
        'Mild','Cool','Mild','Mild','Mild','Hot','Mild'
    ],

    'Humidity': [
        'High','High','High','High','Normal','Normal','Normal',
        'High','Normal','Normal','Normal','High','Normal','High'
    ],

    'Wind': [
        'Weak','Strong','Weak','Weak','Weak','Strong','Strong',
        'Weak','Weak','Weak','Strong','Strong','Weak','Strong'
    ],

    'PlayTennis': [
        'No','No','Yes','Yes','Yes','No','Yes',
        'No','Yes','Yes','Yes','Yes','Yes','No'
    ]
}

df = pd.DataFrame(data)

# Encode categorical values
encoders = {}

for column in df.columns:
```

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```python
# Encode categorical values
encoders = {}

for column in df.columns:
    encoder = LabelEncoder()
    df[column] = encoder.fit_transform(df[column])
    encoders[column] = encoder

X = df.drop('PlayTennis', axis=1)
y = df['PlayTennis']

# ID3
model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)

model.fit(X, y)

# Display tree
plt.figure(figsize=(14, 8))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=['No', 'Yes'],
    filled=True
)

plt.title("ID3 Decision Tree")
plt.show()

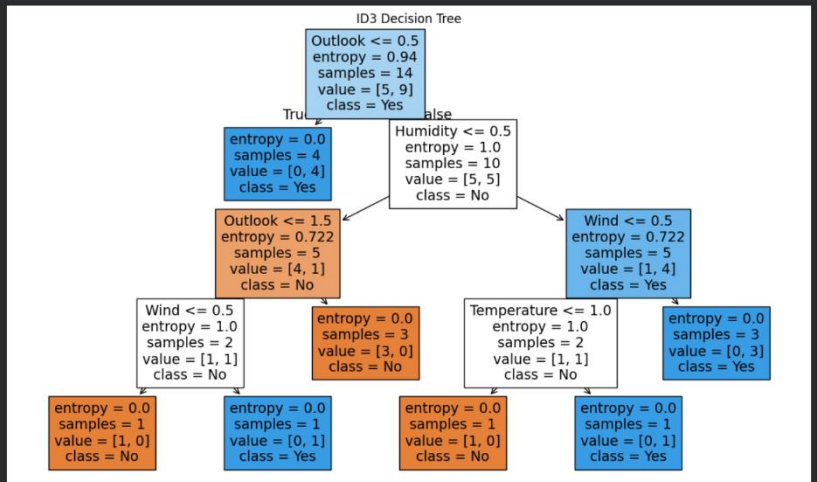
# New sample
sample = [[2, 1, 0, 0]]

prediction = model.predict(sample)

print("Prediction:")
```

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```
    print("No - Do Not Play Tennis")
```




```
Prediction:
No - Do Not Play Tennis
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(
```

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